Rapid Latent Adaptation (RLA) Experimental Protocol

New RIF pipeline for a rigorous online-adaptation experiment on the hydraulic manipulator

Implementation specification for an autonomous coding / experimentation agent
Grounded in the current thesis draft (24 August 2026) and RMA / A-RMA literature

Item	Protocol decision
Primary scientific goal	Show that RLA improves PPO only after the plant is pushed beyond the robustness envelope, while enough actuator headroom remains for a controller to act.
New pipeline	RIF = RLA-conditioned PPO -> frozen ID -> frozen FD -> PhysX. RLA conditions the policy; the primary design removes the legacy small additive ID adapter.
Key prerequisite	Do not train RLA until a privileged/oracle-conditioned PPO demonstrably outperforms a robust no-context PPO on the same stressed plant family.
Online identification claim	Must be a causal chain: plant changes -> estimated latent changes -> PPO action changes -> tracking recovers. A low latent loss alone is not success.
Legacy AAM role	Keep AAM as the thesis evidence for dynamics-relevant latent estimation / ID trustworthiness. Do not treat it as the policy-adaptation mechanism.
Critical confound	Reject adaptation tests dominated by cylinder/action saturation. The old AIF had approximately 92% C1 saturation, 67-82% C2 and 50-57% C4, leaving little corrective authority.

1. Purpose and Scope

This document specifies a new experiment rather than a revision of the legacy AAM chain. The purpose is to create a policy-level rapid latent adaptation pathway that is scientifically capable of demonstrating online adaptation under a condition where the existing PPO controller actually loses robustness. The implementation agent should treat every gate below as binding. A failed gate changes the experiment; it must not be papered over by training a larger estimator.

The desired result is not simply that an estimator predicts a latent. The desired result is that a plant shift which degrades a non-adaptive PPO is detected from causal deployment history, represented by a latent estimate, consumed by the PPO, and followed by a measurable recovery in closed-loop tracking.

plant shift -> z_hat changes -> pi(o, z_hat) changes its action -> tracking recovers

This is intentionally stricter than the previous adaptation argument. It turns the earlier null results into design constraints for the successor experiment.

2. What the Current Thesis Already Establishes

The new pipeline must not repeat work already completed. The 24 August thesis draft establishes the following points, which are treated here as prior evidence rather than hypotheses.

Existing evidence	What it means for RLA
Deployment source	Adaptation data describe the IsaacLab deployment plant, not Simscape. The thesis explicitly uses the command ROM + PhysX as the plant from which adaptation data are generated.
IF transfer	The inverse model transfers from Simscape to IsaacLab with about 84% explained valve-command variance and a small domain spread; it remains frozen downstream.
FD identifiability lesson	A near-perfect one-step forward score once hid almost no valve dependence. Open-loop excitation and input ablation were required to expose and repair the failure.
AAM learnability	AAM can recover the privileged latent well enough that estimated and privileged latent are similar in downstream dynamics-model utility.
AAM information	The deployed two-dimensional AAM latent is dominated by seal condition and pressure information; mass is effectively not decoded.
Task-data limitation	Controller-driven histories do not separate plant identity reliably because command is computed from state and the two are nearly redundant.
Temporal effect	On the dynamics model, allowing the latent to vary online was worth far more than getting the per-episode plant identity correct.
Closed-loop oracle ceiling	Within the legacy AIF architecture, perfect plant knowledge improved lateral error only about 3.4% and cut height about 4.5%, not resolvably better than no plant information.
Robustness envelope	The tested controllers absorbed a 40% valve-effectiveness reduction and even doubled deployment-factor deviations; the experiment did not locate a breakdown point.
Saturation confound	The legacy closed-loop adaptive test ran with very high cylinder saturation, so a bounded additive adapter had little authority exactly where adaptation was supposed to matter.
Policy sensitivity	AIF did consume the latent, but a moving estimate could hurt the feedback loop; filtering the policy latent improved cut-height tracking on the reported checkpoint.

Source: current master thesis draft, especially Sections 4.7-4.10, 5.5-5.8, 7.2-7.12 and 8.1-8.5. These are internal results and should be cited as such in the thesis extension.

3. Literature Basis for the New RLA Pipeline

RLA should follow the control-relevant structure of Rapid Motor Adaptation rather than reusing the system-dynamics latent simply because that latent already exists.

3.1 RMA: policy-trained privileged extrinsics

Kumar et al. (RSS 2021) train RMA in two phases. In Phase 1, a base policy and an environment-factor encoder are trained jointly in simulation: true environment parameters such as mass, center of mass, friction and motor strength are compressed into an extrinsics vector z, and the policy is optimized while receiving z. In Phase 2, an adaptation module estimates z from recent state-action history using supervised on-policy data. Deployment replaces privileged z with the estimate. This distinction is essential: the latent is shaped by the policy objective, not by a separate dynamics-prediction objective [1].

3.2 A-RMA: close the perfect-latent / estimated-latent gap

Kumar et al. (IROS 2022) add a third phase because a policy trained with perfect privileged extrinsics can be brittle when deployment provides an imperfect estimate. They freeze the adaptation module and fine-tune the base policy again with PPO while feeding the estimated extrinsics. The authors report this third phase as critical for reliable deployment [2]. This directly addresses one of the central risks in the hydraulic task.

3.3 RMA for manipulator arms

Liang, Ellis and Henriques (CVPR 2024) apply the same two-stage principle to robotic manipulation: a PPO policy and environment encoder are trained with randomized privileged conditions, then a frozen policy/encoder supervise an adapter from observation/action history using an L2 embedding loss [3]. This is the closest direct precedent for calling the new method an RLA-style manipulator adaptation pipeline.

3.4 Identification and excitation

Classical system-identification literature emphasizes that identifiability is a property of the experiment and input, not of the neural network. Closed-loop feedback can strongly correlate inputs with state, and sufficiently informative excitation is required to distinguish plant dynamics [5,6]. The thesis already observed this failure empirically, so every RLA dataset must be audited for response authority and history informativeness before model training.

3.5 Saturation and authority

Actuator saturation is a structural limit on adaptation, not a small tuning issue. When the plant is saturated, an adaptive correction can lose the authority needed to change the response; classical anti-windup work treats this as a core nonlinear control problem [7]. Consequently, RLA must be evaluated in an "adaptation-limited" region: the baseline degrades because dynamics changed, but the actuators are not pinned at their limits.

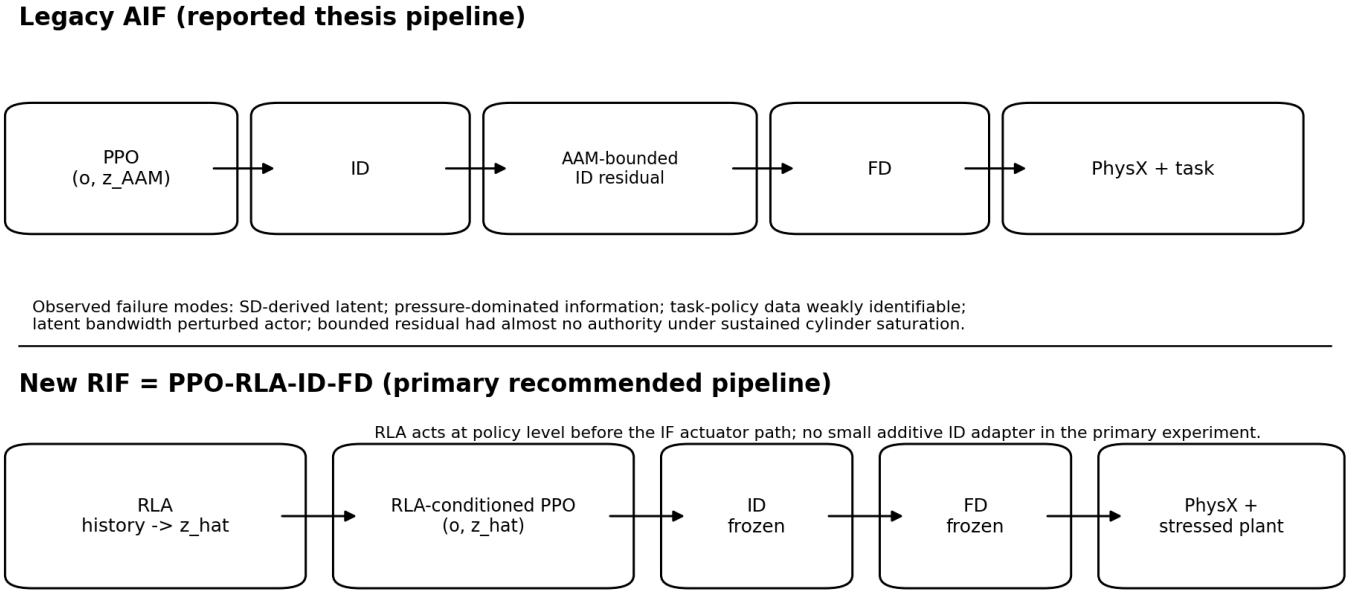


Figure 1. The legacy AIF path and the recommended new RIF path. The new experiment moves adaptation to the policy level and retains the frozen IF actuator chain.

4. Pipeline Definitions and Naming

Stack	Definition
R	PPO -> command-conditioned ROM -> PhysX. Reference deployment representation.
IF	PPO -> frozen inverse dynamics (ID) -> frozen strict forward dynamics (FD) -> PhysX. Robust learned baseline.
AIF	Legacy: AAM-conditioned PPO + bounded latent-conditioned residual around ID -> FD -> PhysX.
RIF	New: RLA-conditioned PPO -> frozen ID -> frozen FD -> PhysX. The primary RLA method.
Oracle RIF	Same RIF policy, but z_hat is replaced by z* from true simulator factors. Training/evaluation only.

Important: RIF is a new pipeline. It is not a renamed AIF. Its latent target, data distribution, PPO training and consumer are different. The old AAM remains a dynamics-model adaptation result; RLA is a policy-adaptation result.

Also important: the current thesis does not prove R and IF are equivalent. It explicitly reports that the two representations induce different control strategies. If equivalence is required for a later claim, it must be tested formally. However, a direct AIF-vs-RIF mechanism comparison should not depend on R = IF because both adaptive methods can be evaluated on the same IF actuator path.

5. Scientific Hypothesis and Success Definition

5.1 Primary hypothesis

$$H_{RLA}: \; E[RIF_{estimated} \mid stressed] \; < \; E[robust\ IF\ PPO \mid stressed]$$

where E is a predeclared closed-loop error metric (primarily lateral tracking and cut-height error), and the stressed region is outside the baseline PPO robustness envelope but inside the actuator-headroom envelope.

5.2 Stronger causal hypothesis

plant change -> z_hat response -> action separation -> post-change recovery

A successful paper/thesis result requires all four links. Any result missing one link is described as latent prediction, context sensitivity, or robustness, not online adaptation.

5.3 What counts as success

Requirement	Interpretation
Baseline degradation	A stress condition causes a material, repeatable loss versus nominal/in-envelope performance.
Headroom retained	The degraded run is not explained by persistent saturation or travel-limit pinning.
Oracle value	Perfect context changes policy actions and recovers a meaningful share of the lost performance.
RLA estimation	z_hat tracks z* sufficiently for the estimated-context policy to retain a substantial fraction of oracle benefit.
Online recovery	After a mid-episode change, RIF_estimated beats frozen-pre-change and mean-context controls in the post-change window.
No privileged leakage	RLA inputs are strictly causal and deployable; true factors appear only as simulation labels.
Matched evaluation	Same plant draws, meshes, trajectory families, switch time and scoring windows across arms.

6. Stage 0: Build the Stress Region Before RLA

The earlier adaptation attempt failed partly because the baseline controller was already robust to the available perturbations. The first new experiment is therefore not RLA training. It is a degradation-envelope sweep.

6.1 Candidate stress factors

Do not automatically reuse all six legacy factors. Admit only factors that satisfy both control relevance and observability. Start from the factor hooks already implemented in IsaacLab, then consider the following in order:

1. Valve/control effectiveness multiplier, including asymmetric per-cylinder effectiveness if already supported. The previous 0.6 multiplier was too mild; sweep further until degradation appears.
2. Payload/end-effector mass and center-of-mass displacement, because these alter the force required for the same motion.
3. Hydraulic response/compliance/flow parameters already implemented in the ROM randomization, but only if the exact RIF path exhibits measurable response authority.
4. Additional untrained response delay or spool-lag perturbation only if it can be implemented as a physically interpretable deployment variation and was not already present throughout training.

Prefer two to four factors. A narrower, control-relevant plant family is stronger than a six-factor family whose members the policy cannot distinguish or does not need to distinguish.

6.2 Factor-admission screen

Gate	Pass condition
F1 response authority	Moving one factor alone changes cylinder response or end-effector response beyond numerical noise.
F2 control relevance	The factor produces a monotonic or at least reproducible degradation trend in the baseline controller.
F3 headroom	The degradation occurs before persistent actuator saturation / lead-cap occupancy dominates.
F4 observability	Recent state-action history changes measurably after the factor changes.
F5 physical validity	No non-finite physics, unrealistic stroke excursions or invalid pressure/force states.

6.3 Proposed headroom gate

The exact thresholds should be reported from the system logs rather than hidden. As a conservative starting rule for this experiment, do not promote a stress point if any adapted cylinder is clipped/saturated for more than roughly 30% of the scored window, if the integrated-target lead cap is continuously active, or if the p01 travel margin falls below the existing 5 mm system gate. The purpose is not to claim 30% is universally safe; it is to avoid repeating the legacy AIF condition in which C1 was saturated about 92% of the scored window.

6.4 Degradation gate

Define the degradation threshold before looking at RLA results. A practical default is a >=25% worsening in a primary tracking metric relative to the same policy on the nominal plant, or a clear failure of one of the pre-

existing physical acceptance gates, reproduced across paired (mesh, trajectory-family) cells. The exact chosen margin must be frozen in the experiment manifest.

7. Stage 1: Formulate the Privileged RLA PPO

This is the key structural change. The privileged latent must be learned for control, not inherited from system dynamics.

7.1 Inputs

$$e_t = [\text{selected normalized plant factors}]$$

The simulator exposes e_t only during training. Use a compact encoder:

$$z^*_t = \mu_\psi(e_t), \quad z^* \in \mathbb{R}^d$$

Recommended initial latent dimension: d = 4. The old d = 2 latent was dominated by one hydraulic direction. If the oracle gate fails because the encoder is demonstrably bottlenecked, validate d = 8 as a single predeclared alternative. Do not tune latent size on the reserved test set.

7.2 Actor

$$a_t \sim \pi_\theta(a \mid o_t, z^*_t)$$

Use the same sole-controller action contract already established in the thesis: the PPO emits the four bounded desired cylinder-motion commands and no DLS/PD controller is present in the scored loop. Keep the same physical observation set o_t used by IF unless a change is explicitly justified and mirrored in the baseline.

7.3 Initialization

Warm-start from the existing verified IF PPO rather than training from scratch. Expand the actor input to accept z, copy the old observation weights, and initialize the new latent-input weights at or near zero so the initial policy reproduces the baseline at z_nom. This preserves the established controller and turns Phase 1 into context-specialization rather than relearning the task.

7.4 PPO objective and safety anchoring

Use the existing PPO implementation and the thesis trust-region/rollback machinery because unconstrained PPO previously improved reward while destroying the actual tracking criterion. However, do not globally anchor the stressed contexts so tightly that the oracle policy is prevented from changing its action. The recommended structure is:

$$L = L_{PPO} + L_{value} - \beta H + \lambda_{nom} \cdot \mathbb{I}(\text{in-envelope}) \cdot KL(\pi_{IF} \,||\, \pi_{RIF})$$

Keep the ordinary update trust region for all data. Apply the stronger reference anchor primarily to nominal/in-envelope samples, preserving nominal competence while allowing context-dependent action separation on stressed plants. Every policy promotion still uses the physical gates, not reward alone.

7.5 Same-budget baseline

Train a robust no-context baseline on the identical stressed plant distribution and budget, or at minimum re-evaluate the frozen IF policy and a matched fine-tuned no-context control. The oracle claim is otherwise confounded by extra optimization rather than privileged context.

8. Stage 2: Oracle Gate - Required Before Training RLA

RLA must not be trained until the downstream consumer proves it can use plant context. Evaluate one fixed privileged policy with multiple latent substitutions.

Arm	Purpose
Oracle z*	True simulator factors passed through the frozen privileged encoder.
Mean z	In-distribution constant carrying no plant information.
Shuffled z	Right latent distribution, wrong plant pairing.
Nominal z	Optional fixed nominal context if it lies in distribution.

8.1 Required oracle evidence

At minimum, oracle must beat mean and shuffled on the selected stressed region, and policy actions must measurably differ. A useful operational target for a strong result is: oracle recovers >=30% of the performance lost by the robust baseline and improves at least one primary metric by >=15% relative to mean context, with a paired bootstrap interval excluding zero. These are proposed design targets, not retrospective thresholds; the agent should freeze them before RLA fitting.

8.2 Action-separation test

$$\Delta_a = \| \mu_\pi(o, z_{oracle}) - \mu_\pi(o, z_{mean}) \|_2$$

Report the normalized action separation and its distribution by cylinder. A context-sensitive policy is required; if the action difference is practically zero, no history estimator can improve the loop even if it predicts z perfectly.

8.3 Stop rule

If the oracle arm does not beat mean/shuffled after one predeclared latent-size alternative and one predeclared stress-family alternative, stop. Report that the tested policy has no useful context-dependent control opportunity in that region. Do not proceed by repeatedly changing metrics or stress ranges until significance appears.

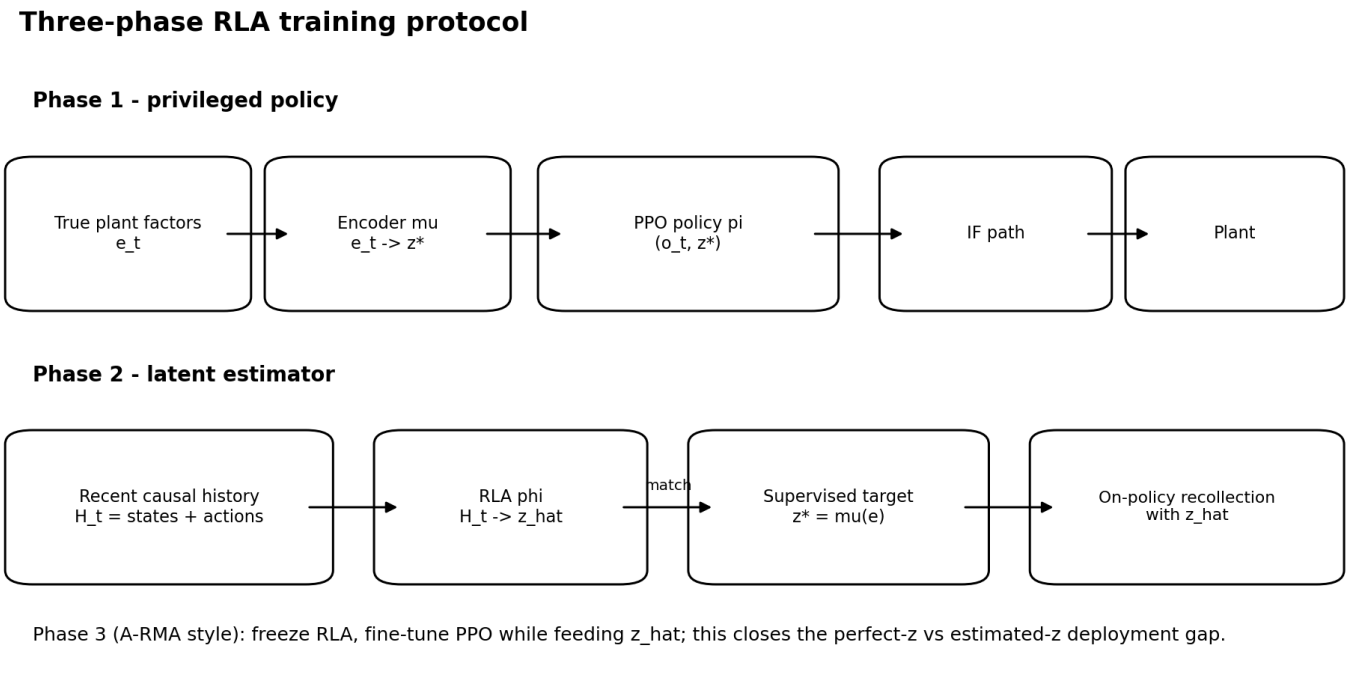


Figure 2. RLA follows the literature structure: privileged control first, history-to-latent estimation second, then an A-RMA-style PPO fine-tune with the imperfect estimator in the loop.

9. Stage 3: RLA Data Generation - New Dataset

This dataset belongs in a new subsection of Chapter 4 (Data Generation and Preparation). It must not reuse the legacy 118-run AAM collection as the primary training set because that collection was built for a different latent and was largely controller-unidentifiable.

9.1 Which plant produces the data?

Primary recommendation: collect the RLA history from the exact RIF deployment path in which the estimator will be used: RLA-conditioned PPO -> ID -> FD -> PhysX, with the selected plant-factor hooks active. True factor values remain available from simulation as labels. This avoids the train/deploy mismatch of training an estimator on ROM histories and deploying it on IF histories.

Before collection, run a factor-response authority check through the exact RIF path. If the existing factor hooks only affect the ROM path and are invisible through IF, do not silently proceed. Either move the factor injection to a shared physical layer that both paths experience, or use the RLA-ROM alternative described in Section 16.

9.2 What is logged?

Group	Signals
Timing	episode_id, env_id, control_step, model_tick, simulation time, switch flag
Proprioception	5 joint positions q, 5 joint velocities qdot
Cylinder response	4 cylinder extensions x and velocities xdot if available at runtime
Hydraulic/model state	4 differential pressures (or chamber pressures if exposed); flag explicitly that these are model states unless measured
Commands	previous applied PPO action(s), current action, pre/post-clipping action, action-limit occupancy
Residual response	existing model-error/residual channels already available to the

	policy, if they are part of the deployable observation contract
Safety/headroom	cylinder stroke margin, command saturation, lead-cap occupancy, safety termination flags
Task evaluation	lateral error, cut-height error, ground contact, head alignment, tool acceleration, reward terms
Privileged labels	active factor vector e_t and z*_t = mu(e_t); never included in RLA input

9.3 RLA input contract

Use only causal signals that exist in deployment. The initial row should be compact and focused on command-response behavior:

$$h_t = [q_t, \dot{q}_t, x_t, \dot{x}_t, \Delta P_t, a_{t-1}, \text{optional model-residual channels}]$$

Do not include true factors, z*, future samples or hidden ROM state. Prefer excluding the desired task reference from the RLA input so the latent is forced to describe plant response rather than trajectory identity. If a reference-derived feature is later added, treat it as an ablation and test whether the latent simply predicts trajectory/time.

9.4 History length and update rate

Start with 0.5 s of history, matching the original RMA principle of rapid inference and shorter than the legacy AAM 1.0 s history plus 250 ms smoother. At the 20 Hz actuator/RLA grid this is 10 rows. Predeclare 1.0 s (20 rows) as the single fallback if the 0.5 s estimator fails. Update RLA at 20 Hz while PPO continues at 60 Hz; PPO consumes the most recent z_hat between RLA updates.

9.5 Three dataset stages

Dataset	Purpose
D0 privileged on-policy	Run the oracle-conditioned policy under randomized/stressed factors. Record state-action histories and target z*. This is the initial RMA-style dataset.
D1 estimated on-policy	Deploy the first RLA so the policy acts on z_hat. Recollect histories produced by its own estimation errors while still labeling them with z*. Append or mix with D0.
D2 switch set	Use controlled mid-episode factor changes. This set is mandatory for adaptation-time evaluation and may also be included in training after its dedicated test subset is reserved.

9.6 On-policy data and the previous identifiability failure

The original RMA method deliberately trains the adapter on policy-generated state/action histories. The hydraulic thesis, however, already showed that ordinary task-controller data can make command and response nearly redundant. Therefore D0 must be audited before RLA training. If history cannot discriminate oracle contexts, do not assume a larger network will fix it.

First try to solve this without adding an external controller: increase diversity of naturally executed trajectories and include the actual previous actions and response residuals. If the task still does not generate identifiable histories, use the fallback in Section 16: a brief, bounded identification maneuver or a reference segment specifically designed to create informative command-response variation. Because the PPO must remain the sole controller in the scored task, any such identification maneuver must either be part of the policy/task contract or be reported as an active-identification variant rather than silently added.

9.7 Splits

Split by complete plant configuration and run, not by windows. Reserve factor keys before training. No history window may cross a run boundary or a factor-switch boundary unless the switch experiment explicitly labels that transition. Store normalizers from the training split only.

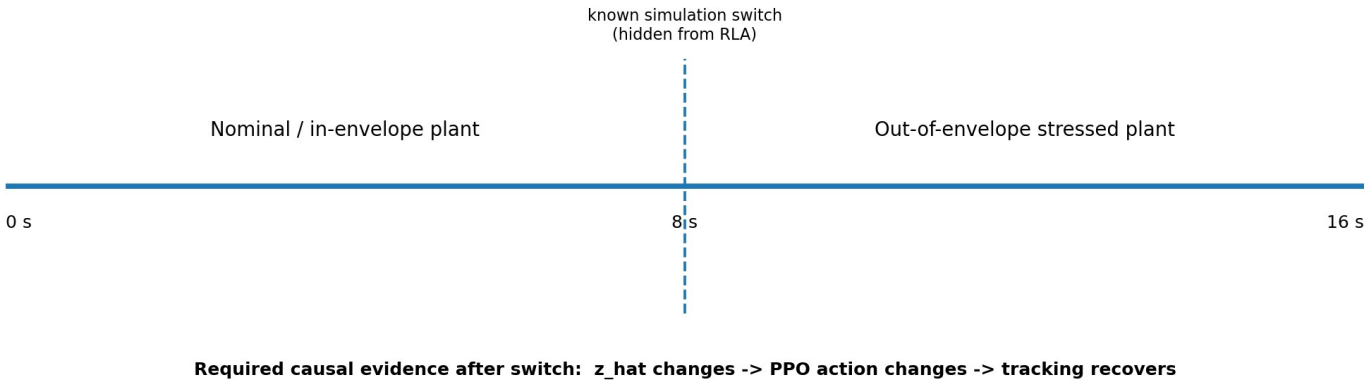


Figure 3. Required mid-episode switch experiment. The change is known to the simulator and hidden from RLA. Success requires a causal latent/action/recovery sequence after the switch.

10. Stage 4: Train the RLA Estimator

10.1 Architecture

Use a modest causal temporal model. A GRU or causal TCN is sufficient; the experiment should not be about architectural search. Recommended initial model: two-layer GRU, hidden size 64-128, final MLP to d=4 latent. Keep parameter count small enough that training is inexpensive and reproducible.

10.2 Loss

The literature baseline is an L2 loss on the privileged embedding. For this thesis, add a policy-aware auxiliary term so an estimator is rewarded for preserving the action-relevant content of z rather than only numerical latent coordinates:

$$L_{RLA} = \lambda_z \|\hat{z} - z^*\|^2 + \lambda_a \|\mu_\pi(o,\hat{z}) - \mu_\pi(o,z^*)\|^2$$

The actor is frozen while computing the auxiliary action term. Start with $\lambda_z = 1$ and a small λ_a (for example 0.1 after normalization); select by validation downstream utility, not by latent RMSE alone. If the simple L2 model already passes all downstream gates, prefer it for parsimony.

10.3 No smoothing by default

Do not reproduce the legacy AAM rolling mean automatically. The old smoother added 250 ms of lag, and the old actor was sensitive to latent bandwidth. Train RLA first without post-hoc smoothing. If estimated z causes action chatter, evaluate a policy-path low-pass filter as a controlled ablation and keep raw z for any model consumer. The filter constant must be selected on validation and the adaptation-latency cost reported.

10.4 Estimator validation

Validation	Why it matters
Latent fit	Normalized RMSE/cosine or correlation to z* on unseen plant keys.
Factor decode (diagnostic)	Linear or small decoder from z_hat to selected physical factors. This is diagnostic only; RLA identifies extrinsics, not necessarily named parameters.
Action fidelity	Difference between actions under z_hat and z* on the same observations.
Oracle-retention	Fraction of oracle closed-loop advantage retained by z_hat.
Switch latency	T50/T90 of z_hat response to the known factor switch.

A low latent loss is never the promotion criterion by itself.

11. Stage 5: A-RMA-Style PPO Fine-Tuning with RLA in the Loop

After RLA is competent, freeze it and fine-tune the RIF PPO while feeding z_hat rather than z*. This directly follows A-RMA and is the main defense against the perfect-latent / estimated-latent distribution gap.

1. Freeze the RLA weights and privileged encoder.
2. Collect PPO rollouts using z_hat from current causal history.
3. Keep the same stress family used for the oracle policy, with the same in-envelope samples mixed in to preserve nominal behavior.
4. Use the existing PPO trust-region / rollback checks and physical promotion gates.
5. Optionally inject small validation-calibrated latent corruption during the privileged-to-estimated transition if the policy overfits to precise z*; report it explicitly.
6. Do not alter the ID or FD weights during this phase. The experiment is about policy adaptation.

12. Saturation: Mandatory Confound Control

The previous AIF test cannot be used as evidence that online policy adaptation is impossible because the downstream additive adapter had almost no command authority. The new experiment must deliberately separate robustness degradation from actuator impossibility.

12.1 Preflight every selected stress point

Diagnostic	Required reporting
Command saturation occupancy	Per cylinder, pre- and post-switch
Integrated-target lead-cap occupancy	Per cylinder
Travel margin	Minimum and p01
Action clipping	Fraction of actor outputs clipped
Desired vs achieved cylinder rate	Detect rate-limit pinning
Control-effort change	Confirm RIF actually uses additional/different authority

12.2 Reject the wrong kind of degradation

If baseline tracking degrades only when the plant is physically incapable of following the command, RLA cannot be credited or blamed. Move the operating point: reduce path speed/amplitude, choose a factor that changes

response before full saturation, or use a different trajectory family. The target regime is: baseline dynamics mismatch causes error, but an oracle policy can still correct it. The oracle gate proves that headroom exists in a control-relevant sense.

13. Final Experimental Matrix

All primary adaptation claims should use within-checkpoint or matched-policy substitutions wherever possible.

Arm	Context	Role
IF robust PPO	No latent	Robustness baseline
RIF mean	Mean latent	In-distribution no-information control
RIF shuffled	Wrong latent	Controls for extra dimensions / latent distribution
RIF oracle	z* from true factors	Upper bound and positive control
RIF frozen pre-change	Latch z_hat before switch	Tests whether online updating itself matters
RIF estimated	Online z_hat	Proposed method
AIF legacy	AAM + bounded ID adapter	Historical adaptation mechanism; secondary comparison
R reference	ROM plant controller	Representation/deployment reference, not an automatic equivalent control

13.1 Primary comparisons

1. RIF oracle vs RIF mean/shuffled: does plant context have value?
2. RIF estimated vs RIF frozen-pre-change: does online re-identification improve recovery after the same switch?
3. RIF estimated vs IF robust PPO: does the complete adaptive method improve the PPO outside its robustness envelope?
4. RIF estimated vs RIF oracle: how much of the available adaptation ceiling is retained?

13.2 AIF vs RIF

Treat AIF-vs-RIF as a secondary architectural comparison unless both are retrained under matched data, stress distribution, seeds, PPO budget and evaluation. The legacy AIF checkpoint was trained under a different run configuration, and the thesis already warns against attributing cross-stack differences to one component. The strongest causal RLA claim therefore comes from RIF internal ablations, not from simply placing AIF and RIF in one table.

If time permits, create a matched AIF reproduction under the same stress family and evaluation. Then interpret the difference mechanistically: AIF uses an SD-derived latent and a small ID residual; RIF uses a policy-trained latent and policy-level action adaptation. This is a legitimate comparison of two complete adaptation mechanisms, not a pure estimator comparison.

14. Equivalence of R and IF: How to Handle It Correctly

The current thesis does not establish R = IF. A non-significant difference is not evidence of equivalence. If a statement of practical equivalence is needed, run a predeclared equivalence test on the selected stress region.

14.1 Suggested equivalence margins

Choose margins from engineering significance, not from observed results. A defensible starting proposal is: primary tracking metrics within +/-10% of the reference mean, ground-contact fraction within +/-2 percentage points, and no material difference in safety-gate violations. Use paired confidence intervals / TOST-style reasoning over the same (mesh, family, plant) cells. Freeze the margins before scoring.

14.2 If equivalence passes

Then R and IF may be described as practically equivalent for the selected experiment and operating region only. That supports generalizing the adaptation result across the two actuator representations more confidently.

14.3 If equivalence fails

Nothing fundamental breaks. Keep R as the deployment/reference representation and conduct AIF-vs-RIF on the shared IF path. Do not use R to claim causal superiority of one adaptation mechanism over another.

15. Statistical and Reporting Protocol

1. Use paired (mesh, trajectory-family, plant-draw) cells as the closed-loop replication unit. Do not count deterministic evaluation seeds as independent replications.
2. Fix plant draws, switch times, trajectory families, meshes, scoring horizon and metrics before final scoring.
3. Use paired bootstrap intervals for effect sizes and report sign counts. For equivalence, use predeclared practical margins and an equivalence procedure rather than interpreting $p > 0.05$ as equality.
4. Report median as well as mean when a small number of failure cells can carry the mean.
5. Score reserved plant keys once, after architecture and history length are fixed.
6. Every plot showing recovery must overlay the switch time and include baseline, oracle, frozen and estimated arms.
7. Every adaptation table must include saturation/headroom diagnostics so improved or failed tracking cannot be attributed to missing actuator authority.

16. Failure Branches and Predeclared Alternatives

The experiment is designed to be finishable under a short thesis deadline. Each branch below is an alternative, not an invitation to unlimited tuning.

Failure	Predeclared response
A. Baseline never degrades	Increase severity along the same physical factor direction until a clear degradation appears, but stop before persistent saturation. If no such region exists, switch to a factor that acts faster than the feedback loop (e.g. additional response lag or asymmetric effectiveness) or a slightly different feasible trajectory demand.
B. Degradation appears only with saturation	Reject that region. Reduce task demand or choose another factor. Do not claim RLA failed when the plant is pinned.
C. Oracle PPO has no advantage	Try one predeclared alternative stress family and one latent-dimension alternative. If still no oracle gap, terminate RLA: the controller has no useful context-conditioned action opportunity.
D. Oracle works, RLA fit is poor	Increase history from 0.5 to 1.0 s; add causal response residuals; use a GRU/TCN; collect estimated-policy on-policy D1. Do not add privileged inputs.

E. RLA fits z but closed-loop degrades	Run A-RMA Phase 3; train with estimated z; test policy-path filtering; add action-fidelity loss. This is the expected perfect-vs-estimated latent gap.
F. Task-only histories remain unidentifiable	Use a brief, explicitly declared active-identification maneuver/reference segment or safe excitation window. Keep it inside the action contract and report it as an active-identification variant. Do not hide an external controller inside RLA.
G. IF path cannot express factor variation	Move variation to a shared physical layer or run an RLA-ROM variant. Do not train an estimator against labels that do not affect its actual plant.
H. Time is running out	Stop after the oracle gate. A positive oracle gap with a failed estimator is still a clear future-work result; a negative oracle gate is a scientifically valid reason not to build RLA.

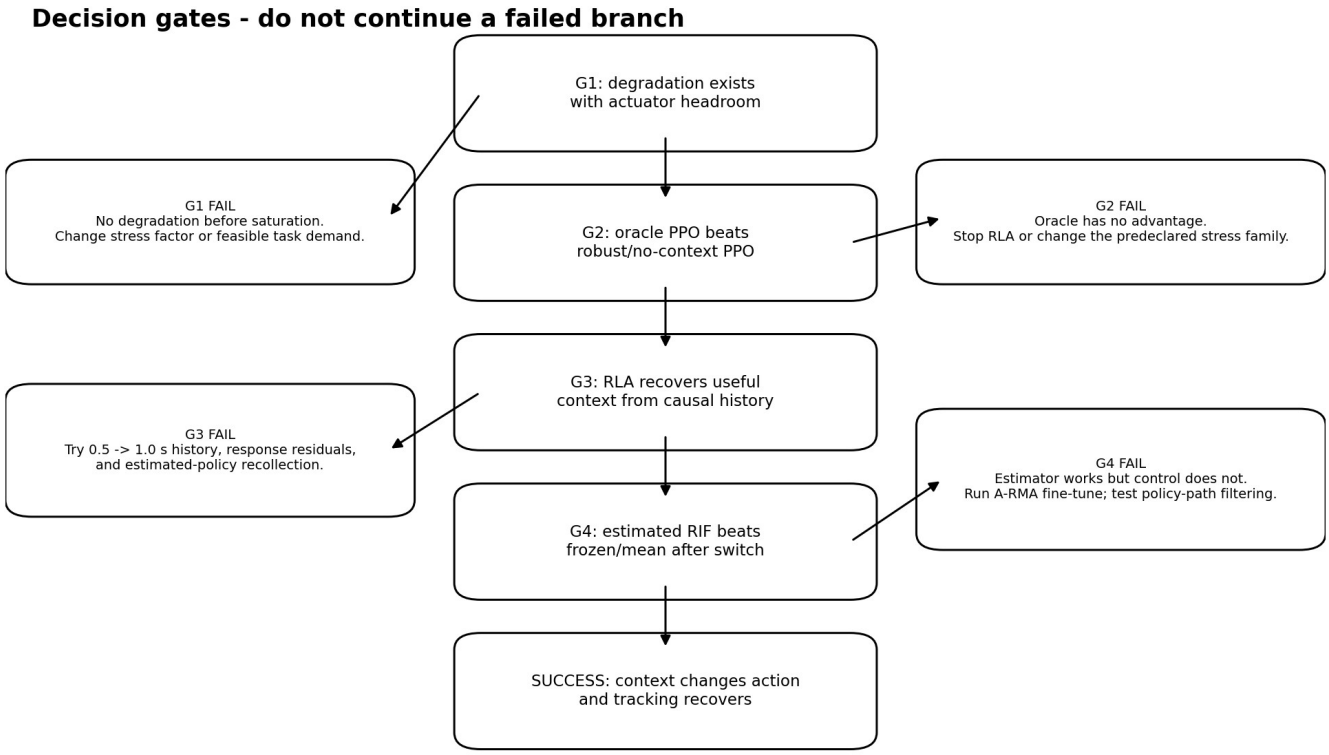


Figure 4. Gate-based decision tree. A failed upstream condition changes the experiment; downstream training is not used to compensate for a missing adaptation opportunity.

17. Why AIF Did Not Work, and Why RIF Is a Different Test

Legacy AIF limitation	**RIF design response**
Latent learned for SD prediction, with strongest gain in pressure rather than controller-integrated extension/velocity.	Latent is learned jointly with the PPO reward/return so its semantics are control relevant by construction.
Task-controller collection made plant identity weakly identifiable.	RLA uses explicit on-policy history audits and estimated-policy recollection; a declared active-identification fallback exists if natural motion is insufficient.
Two-dimensional latent largely represented seal condition; mass was not recovered.	Start from factors admitted by policy degradation/authority screens; use d=4 with one d=8 fallback instead of inheriting d=2.
Oracle plant knowledge had only a 3.4-4.5% ceiling in the old architecture.	Oracle gate is run before estimator training in the new architecture and stressed region; no RLA work continues without a useful oracle gap.
Bounded ID residual capped at 0.15 while multiple cylinders were saturated for most of the scored window.	Primary RIF adapts at policy level and rejects stress points dominated by saturation. No tiny additive actuator correction is

	required.
Moving latent disturbed the feedback policy; legacy smoothing added delay.	A-RMA-style fine-tuning trains PPO on estimated z; policy-path filtering is a controlled fallback, not an automatic patch.
Baseline already saw model-error channels that carried plant consequences.	Keep those channels for fairness; RLA must prove additional value beyond consequence-based feedback, primarily in abrupt/out-of-envelope conditions.

This is the core thesis narrative if RIF succeeds: AIF did not fail because "AAM was inaccurate." The thesis already shows the estimator was not the main limiting component. AIF failed because its latent was optimized for a different consumer, the available plant variation did not exceed PPO robustness, and the actuator-level correction had little authority. RIF changes those exact conditions deliberately.

18. Required Online Identification Evidence

To use the phrase "true online identification through latent identification," the thesis should separate three levels of evidence.

Evidence level	**Required result**
Level 1 - latent inference	z_hat predicts z* on unseen plant keys.
Level 2 - online re-identification	After a mid-episode plant switch, z_hat moves toward the new z* with measured T50/T90 and does not show the same movement in the sham arm.
Level 3 - control consequence	The latent change causes a different PPO action and recovers task performance relative to a frozen-pre-change latent.

Only Level 3 supports the main online-adaptation claim. If physical-factor decoding from z_hat is weak but Level 3 is strong, describe RLA as identifying a control-relevant latent extrinsic rather than explicitly identifying named hydraulic parameters. This is consistent with RMA, whose extrinsics need not correspond one-to-one with physical parameters.

19. Proposed Quantitative Promotion Gates

The following are recommended targets for a strong experiment. They should be finalized before running the reserved test set and can be adjusted only with a written engineering rationale based on known system gates, not on RLA outcomes.

Gate	**Target**
G0 - physical validity	No non-finite physics; factor hooks affect the exact RIF path; run remains inside travel/safety limits.
G1 - adaptation region	Baseline primary error >=25% worse than nominal or fails a pre-existing physical gate, while per-cylinder saturation is not persistently dominant (target <=30% occupancy).
G2 - oracle opportunity	Oracle recovers >=30% of lost performance and improves >=15% vs mean on >=1 primary metric with paired interval excluding zero.
G3 - policy sensitivity	Oracle-vs-mean action separation is nontrivial and repeatable across plant cells/cylinders.
G4 - RLA utility	Estimated latent retains >=60% of oracle improvement on validation.
G5 - online speed	On switch tests, target T50 <=0.5 s and T90 <=1.0 s, unless plant observability itself is slower and that delay is measured.
G6 - adaptation claim	RIF estimated beats frozen-pre-change and mean on the post-switch primary metric, with paired uncertainty and consistent sign.

G7 - nominal retention	RIF does not materially degrade nominal/in-envelope performance or safety relative to IF beyond the predeclared tolerance.

20. What Must Be Created First - Agent Build Order

1. A read-only experiment manifest that records the baseline IF checkpoint, current action/observation contract, factor hooks, rate limits, saturation definitions, meshes, trajectory families and physical gates.
2. A stress-sweep runner that can scale one factor or a vector of factors without training any policy.
3. A saturation/headroom report generated for every stress point.
4. A privileged factor encoder + expanded PPO actor that can consume z*, initialized from the verified IF policy.
5. A same-budget no-context PPO control for the stressed distribution, if the frozen IF checkpoint is not an adequate matched baseline.
6. The oracle / mean / shuffled substitution harness and action-separation logger.
7. Only after the oracle gate passes: the RLA data recorder, with strict causal masks and run-level splits.
8. RLA estimator training code and validation report.
9. Estimated-policy recollection loop D1 and A-RMA Phase 3 PPO fine-tune.
10. Mid-episode switch evaluator with frozen, mean, shuffled, oracle and online arms.
11. A matched AIF evaluation on the same stress/switch cells as a secondary comparison.

21. File / Artifact Contract for the Implementation Agent

Use project-consistent paths, but create logically equivalent artifacts with names that make the phase explicit. The agent should not overwrite existing thesis experiments.

Artifact	**Contents**
configs/rla_stress_sweep.*	Factor ranges, severity grid, headroom thresholds
configs/rla_phase1_privileged.*	Privileged PPO/encoder settings
configs/rla_phase2_estimator.*	History contract, latent dimension, estimator settings
configs/rla_phase3_finetune.*	Estimated-latent PPO fine-tune
experiments/RLA/phase1/...	Privileged PPO checkpoints and oracle gate results
experiments/RLA/data/D0...	Privileged on-policy histories
experiments/RLA/data/D1...	Estimated-policy on-policy histories
experiments/RLA/data/D2_switch...	Switch trajectories
experiments/RLA/phase2/...	RLA checkpoints / validation
experiments/RLA/phase3/...	Final RIF checkpoints
experiments/RLA/eval/...	Paired final tables, traces, bootstraps, saturation reports
experiments/RLA/manifest.json	Checkpoint hashes, git commit, seeds, factor keys, exact commands

Every saved result should carry the source checkpoint hash, git commit, factor key, plant severity, mesh/family, scoring window and whether z was oracle/estimated/frozen/mean/shuffled. A result without provenance is not eligible for the thesis.

22. Thesis Integration Plan

22.1 Chapter 4 - Data Generation and Preparation

Add a new subsection titled approximately "RLA Policy-Adaptation Collection." It should state: (i) why the old AAM collection is not reused as the primary dataset, (ii) the exact RIF plant/path that generates histories, (iii) admitted stress factors and severity region, (iv) D0/D1/D2 collection phases, (v) causal input contract, (vi) history length and rates, (vii) split by plant key/run, and (viii) identifiability/headroom audits.

22.2 Chapter 5 - Learned Models and Online Adaptation

Add a new subsection "Rapid Latent Adaptation (RLA)." Put the privileged encoder, RLA architecture, L2/action-fidelity loss, on-policy recollection, switch-latency metrics and A-RMA-style policy fine-tuning here. Keep AAM in its current role and explicitly say RLA is a new policy-specific latent, not a continuation of the SD latent.

22.3 Chapter 6 - Control Framework

Add RIF as a fourth stack or successor experimental stack: RLA-conditioned PPO -> frozen ID -> frozen FD -> PhysX. State that no classical controller is present at runtime and that the primary RIF method has no additive inverse-model adapter.

22.4 Chapter 7 - Experimental Evaluation

Add the experiments in this order: degradation envelope/headroom; oracle gate; RLA estimation; estimated-policy fine-tune; mid-episode switch; robustness recovery curve; AIF-vs-RIF secondary comparison; R/IF equivalence only if formally tested. This order mirrors the causal logic and makes a failed upstream condition visible.

22.5 Chapter 8 - Interpretation

If RIF succeeds, the conclusion should not say that AAM was wrong. Say that the legacy pipeline established model-side latent utility but had a low control-side oracle ceiling and inadequate actuator authority. RIF succeeds because the latent is trained for the PPO objective, the stress is outside the robustness envelope, history is collected on-policy for the deployed stack, imperfect-latent fine-tuning is included, and the evaluation is run with actuator headroom.

23. Minimal Strong Experiment if Time Is Extremely Limited

If time permits only one decisive result, prioritize the following sequence and stop anywhere a gate fails:

1. Find one stress factor and severity where IF PPO degrades >=25% without persistent saturation.
2. Train one d=4 privileged RIF PPO from the IF checkpoint.
3. Show oracle > mean/shuffled and action separation.
4. Collect D0 and train one 0.5 s RLA estimator.
5. Run one A-RMA-style fine-tune block with frozen RLA.

6. Evaluate one mid-episode switch across the existing 33 paired cells: oracle, estimated, frozen-pre-change, mean, IF baseline.

This experiment alone can support the central claim if the causal chain resolves. Everything else is reinforcement or comparison.

24. Agent Stop Conditions

Condition	Decision
Stop before RLA	No stress point produces degradation with headroom.
Stop before RLA	Oracle context does not improve the stressed PPO.
Stop estimator tuning	RLA cannot infer useful z after the one predeclared history-length fallback and on-policy recollection.
Stop adaptive claim	Estimated z moves but action does not; or action moves but tracking does not recover.
Reject result	Improvement appears only in cells dominated by saturation, infeasible references or safety termination.
Reject comparison	AIF/RIF were evaluated on different plant draws, meshes, scoring windows or checkpoints without a paired design.

25. References

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[2] A. Kumar, Z. Li, J. Zeng, D. Pathak, K. Sreenath, and J. Malik, "Adapting Rapid Motor Adaptation for Bipedal Robots," IEEE/RSJ IROS, 2022. arXiv:2205.15299.

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[8] K. K. Murera, "Learning-Based Online Adaptation for Control of a Hydraulic Manipulator," Master Thesis draft, Robotics Research Lab, RPTU Kaiserslautern-Landau, 24 August 2026. Internal source used here: Chapters 4-8.

26. Final Instruction to the Implementation Agent

Do not begin by training an estimator. Begin by proving there is a recoverable control opportunity. The final experiment is valid only if the baseline is demonstrably outside its robustness envelope, the actuator still has

headroom, a privileged policy can exploit context, the deployable RLA recovers that context from causal history, and the resulting action change improves the closed loop. The previous AAM result is not discarded; it becomes the negative/diagnostic predecessor that explains why the new pipeline is designed differently.

If any upstream condition fails, record the failure and follow the declared alternative. Do not change the success metric after seeing the result. Do not use reserved test plants for model selection. Do not hide privileged information in RLA inputs. Do not attribute an effect to adaptation when actuator saturation or a different training run can explain it.